

28 Oct 25

0 0 0 0	0
0 0 0 1	1
0 0 1 0	2
0 0 1 1	3
0 1 0 0	4
0 1 0 1	5
0 1 1 0	6
0 1 1 1	7
1 0 0 0	8
1 0 0 1	9

LSB سے گراؤ یا وہ same رہتا یا بڑھتا یا چھوٹا ہے

ہو سکتا

MSB سے گراؤ یا وہ same رہتا یا بڑھتا یا چھوٹا ہے لیکن
same رہنے کی guarantee نہیں ہے

Interrupt Service Routine

↳ once completed go back to original work/task

ARM gives 8 bits for priority
i.e. 0 to 255 values

↓ but Tiva says i need only
3 bits i.e. 0 to 7

↓ but takes the three bits of
MSB and leaves the three
bits of LSB (i.e. drops them)
62500000

→ smallest no has the highest priority
i.e. 0

[7:5] bits are used

Exception → problem during program working

Interrupts → By Peripherals eg UART, SPI



Processor → deal with interrupts at global level and deal priority

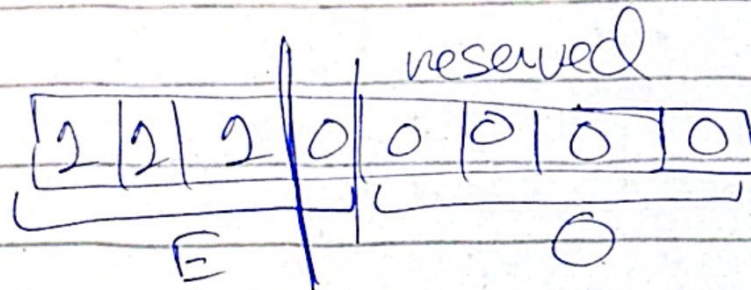


NVIC → from which peripheral to take, enable/disable and give priority



Peripheral → from which pin to take and how (level (high, low)) or edge (rising/falling/both)

Context M4 - Generic User guide $\rightarrow$ pg 20



$$7 \times 2 = 14 \Rightarrow E$$

so

0xEE0

① Processor Level Configuration

① Enable/Disable $\left\{ \begin{array}{l} \text{PRIMASK} \\ \text{--- 3, 2, 1} \\ \text{FAULTMASK} \\ \text{--- 2, 1} \end{array} \right.$

1 $\Rightarrow$ Disable

0 $\Rightarrow$ Enable

② ~~BASEPRT~~ Priority — BASEPRI

$$\begin{array}{r} \overset{4}{32} \overline{) 1387} \text{ no of interrupts} \\ \underline{128} \\ 10 \end{array}$$

→ 4 whole registers and first 10 bits of S reserved

→ Interrupt no 32 reserved

$$\begin{array}{r} \overset{1 \Rightarrow EN1}{32} \overline{) 32} \\ \underline{32} \\ 0 \end{array}$$

0 → Bit 0

② NVIC

0xE000E000

Enable — Quotient ENQ

Remainder — Bit no

Interrupt
no / 32

Disable — Quotient DISQ

Remainder — Bit no

8 bits used to
set priority
of interrupt

PRI

$$\begin{array}{r} 8 \overline{) 32} \\ \underline{32} \\ 0 \end{array}$$

com feed 4 interrupts
in one register

$$\begin{array}{r} 34 \\ 4 \overline{) 136} \\ \underline{136} \\ 0 \end{array}$$

So • PRI 1 to PRI 35

→ if interrupt 33

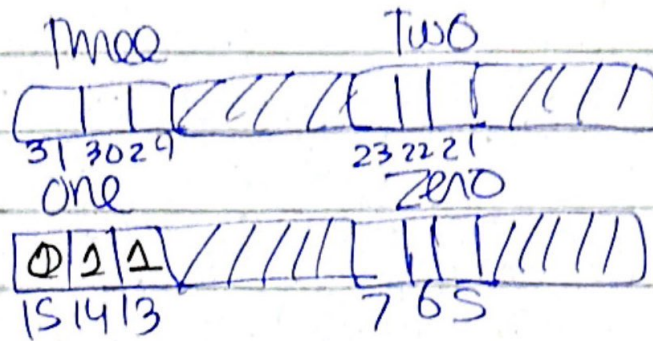
$$\begin{array}{r} 8 \overline{) 33} \\ \underline{32} \\ 1 \end{array}$$

→ Priority register 8

So

— PRI — Quotient PRIQ
Interrupt/4

Dividing by 9's max remainder 3



so 60 ^{7ans} ~~sw~~ multiply ≈ 2 ~~5~~ priority 3.

0x00006000
 interrupt interrupt interrupt interrupt
 D C B A

③ Peripheral level Configuration

